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Plasma gasification for conversion of waste into energy



Fig. 1: A plasma torch in operation

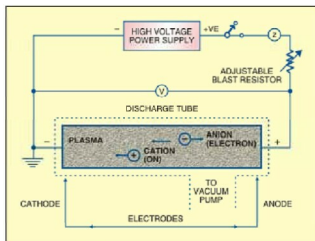


Fig. 2: Simplified circuit of a plasma torch

of all matter is in the plasma state.

Plasmas are electrically conductive. These consist of freely moving charged particles, i.e., electrons and ions along with neutrals—all at different temperatures. Plasma densities vary widely from 10^3 to 10^{23} charged

particles/ m^3 while the corresponding temperatures vary from 10^3 Kelvin to 10^8 Kelvin. For details, you may refer the chart at '<http://FusEdWeb.Ilnl.gov/CPEP/>'

What a plasma gasification system can do?

It can completely dissociate all organic and inorganic matter into their elemental compounds for recovery and recycling. Plasma can be industrially produced using a plasma torch (Fig. 1). The plasma torch can produce extremely high temperatures that cannot be otherwise created except through nuclear fusion/fission.

By passing a DC current between the cathode and the anode of the plasma arc torch and simultaneously passing of air in the annular space of the torch, an extremely high-temperature environment is created ranging in temperature from 5000°C to $10,000^\circ\text{C}$. A simplified circuit of the plasma torch is shown in Fig. 2.

A plasma reactor operates in an oxygen-starved environment and hence there is no combustion. With core temperature running up to $10,000^\circ\text{C}$, plasma is able to break down toxic compounds within milliseconds, avoiding the formation of secondary